MILITARY SPECIFICATION

DRIVE ASSEMBLY, GUN FOR ARMAMENT SUBSYSTEM, HELICOPTER, 20MM: M97E1 AND M97E2

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements in addition to those on the drawings. It provides quality assurance provisions for the fabrication of parts, assembling and packaging for the Drive Assembly, Gun for Armament Subsystem Helicopter, 20MM: M97E1 and M97E2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-6021	-	Castings, Classification and Inspection of.
MIL-I-6868	-	Inspection Process, Magnetic Particle.
MIL-M-8609	-	Motors, Direct-current, 28-Volt System, Aircraft, General Specification for.
MIL-S-45743	-	Soldering, Manual Types, High Reliability Electrical and Electronic Equipment.
MIL-W-63150	-	Weapons and Support Materiel, Standard Quality Assurance Provisions for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-410	-	Nondestructive Testing Personnel Qualification and Certification.
MIL-STD-810	-	Environmental Test Methods.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT

189F734 - Drive Assy, Gun.

PACKAGING DATA SHEET

189F734 - Drive Assy, Gun.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18	-	Method of Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM B 117	-	Method of Salt Spray (Fog) Testing.

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications and standards.

3.2 Parts. All parts shall comply with applicable drawings, specifications and standards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

3.3 Gun drive assembly. The Gun Drive Assembly shall comply with all requirements specified on Drawing (Dwg.) F 189F734, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3.1 Performance characteristics.

3.3.1.1 Acceleration. Under no-load condition when the input voltage of 28 + 0.5 Vdc is applied to Pins B and C of the drive motor, the output gear shall rotate counterclockwise a minimum of 12 revolutions within the first second when viewed from the output end of the drive assembly.

3.3.1.2 Steady-state operation. With 28 + 0.5 Vdc and a torque of 68.5 inch-pounds applied, the drive assembly output gear shall rotate counterclockwise at the steady state operation of 817 to 876 revolutions per minute (rpm) when viewed from the output end of the drive assembly. Steady-state operation shall be reached within one second.

3.3.1.3 Maximum torque. With power applied as specified in 3.3.1.1, mechanical disconnection from the drive motor, caused by the shearing of pins, shall not occur when a maximum torque of 800 inch-pounds is applied to the output gear.

3.3.1.4 Environmental characteristics (first article inspection only).

3.3.1.4.1 Temperature altitude. Drive assemblies shall operate while exposed to conditions specified in Method 504.1 of MIL-STD-810 simulating on altitude of 20,000 feet (temperature of -25°F and barometric pressure 2.12 inches mercury) while energized with line-to-neutral voltage of 124 volts. The test shall be performed as specified in 4.5.6.1.1.

3.3.1.4.2 Vibration. Drive assemblies shall withstand a vibration of 80 Hz with the motor functioning at 6200 rpm for 400 duty cycles. A duty cycle shall be defined as time-on for 3 seconds and time-off for 5 seconds. After each series (5 duty cycles), the motor shall be stopped and cooled to ambient temperature. Cleaning and lubrication shall be accomplished after each 2 series. The magnitude and direction of vibration shall be resultant of 0.15 inch total excursion parallel to the drive rotor axis and 0.40 inch total excursion perpendicular to the drive rotor axis. In addition, with the motor stopped, the drive shall withstand vibration of 80 Hz at 0.40 inch total excursion in a direction parallel to the drive rotor axis for a period of one (1) hour. The amplitude of vibration shall be within 10 percent of the specified values. The test shall be performed as specified in 4.5.6.1.2.

3.3.1.5 Proof test (first article inspection only). Gun drive assemblies shall operate when assembled to an accepted Universal Turret Subsystem (UTS). The test shall be performed as specified in 4.5.6.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article by the contractor shall be as specified in the contract.

3.5 Interchangeability. Unless otherwise specified on the applicable drawings, all parts having identical part numbers shall be functionally and dimensionally interchangeable.

3.6 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. The parts shall be clean and free of burrs, sharp edges, cracks, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required markings and stamping shall be neat and sharply defined. (Required packing shall be dry).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-W-63150 shall apply and are hereby made a part of this detail specification.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted. (see 6.2)

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Gearshaft	D101D4498	5
Gear, spur	C116C3126	5
Cover, end	D117D2954	5
Bearing Ball Annular	C11830451-1	5
Bearing Ball Annular	C11830451-3	5
Gear, Spur	C125C2691	5
Gear, Spur	C125C2692	5
Gear, Pinion	C125C2729	5
Spacer, sleeve	B135B5773-25	5
Spacer, sleeve	B135B5773-26	5
Spacer, sleeve	B135B5773-27	5
Shaft	B135B5783	5
Retainer	B135B5785	5
Bearing, Needle	B135B5970-4	5
Pin	B135B6695	5
Housing, Drive	F189F727	5
Motor, Elec, Drive	D905D490	5
Gun Drive Assy	F189F734	5

4.3.2 Inspections to be performed. See Table I specified herein.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen in the sample to comply with any of the stated requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE		SHEET 1 OF 5		DRAWING NUMBER
	Drive Assy, Gun and components				See Below
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Gearshaft (Dwg. D10LD4498)				
	Examination for Defects	5		3.2	4.4.2.1
	Magnetic Particle Inspection	(a)		3.2	4.5.3
	Hardness	(a)		3.2	4.5.2
	Salt Spray Test	(a)		3.2	4.5.1
	Test Load on Welded Joint	(a)		3.2	4.5.4
	Gear, Spur (Dwg. C116C3126)				
	Examination for Defects	5		3.2	4.4.2.2
	Magnetic Particle Inspection	(a)		3.2	4.5.3
	Hardness	(a)		3.2	4.3.2
	Salt Spray Test	(a)		3.2	4.5.1
	Cover, End (Dwg. D117D2954)				
	Examination for Defects	5		3.2	4.4.2.3
	Salt Spray Test	(a)		3.2	4.5.1
	Bearing, Ball Annular (Dwg. C1183045-1)				
	Examination for Defects	5		3.2	4.4.2.4
	Hardness	(a)		3.2	4.5.2
	Bearing Stability	(a)		3.2	4.5.5

NOTES: (a) This test shall be performed on the 5 samples subjected to the examination for defects.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE		SHEET 2 OF 5		DRAWING NUMBER
cont'd	Drive Assy, Gun and components				See Below
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Bearing, Ball, Annular (Dwg. C11830451-3)				
	Examination for Defects	5		3.2	4.4.2.5
	Hardness	(a)		3.2	4.5.2
	Bearing Stability	(a)		3.2	4.5.5
	Gear, Spur (Dwg. C125C2691)				
	Examination for Defects	5		3.2	4.4.2.6
	Magnetic Particle Inspection	(a)		3.2	4.5.3
	Hardness	(a)		3.2	4.5.2
	Salt Spray Test	(a)		3.2	4.5.1
	Gear, Spur (Dwg. C125C2692)				
	Examination for Defects	5		3.2	4.4.2.7
	Magnetic Particle Inspection	(a)		3.2	4.5.3
	Hardness	(a)		3.2	4.5.2
	Salt Spray Test	(a)		3.2	4.5.1
	Gear, Pinion (Dwg. C125C2729)				
	Examination for Defects	5		3.2	4.4.2.8
	Magnetic Particle Inspection	(a)		3.2	4.5.3
	Hardness	(a)		3.2	4.5.2
	Salt Spray Test	(a)		3.2	4.5.1

NOTES:

(a) This test shall be performed on the 5 samples subjected to the examination for defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

MIL-D-63516(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 5		DRAWING NUMBER See Below
cont'd	Drive Assy, Gun and components				NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spacer, Sleeve (Dwg. B135B5773-25)				
	Examination for Defects	5		3.2	4.4.2.9
	Salt Spray Test	(a)		3.2	4.5.1
	Spacer, Sleeve (Dwg. B135B5773-26)				
	Examination for Defects	5		3.2	4.4.2.10
	Salt Spray Test	(a)		3.2	4.5.1
	Spacer, Sleeve (Dwg. B135B5773-27)				
	Examination for Defects	5		3.2	4.4.2.11
	Salt Spray Test	(a)		3.2	4.5.1
	Shaft (Dwg. B135B5783)				
	Examination for Defects	5		3.2	4.4.2.12
	Hardness	(a)		3.2	4.5.2
	Salt Spray Test	(a)		3.2	4.5.1
	Retainer (Dwg. B135B5785)				
	Examination for Defects	5		3.2	4.4.2.13
	Salt Spray Test	(a)		3.2	4.5.1

NOTES:

(a) This test shall be performed on the 5 samples subjected to the examination for defects.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE		SHEET 4 OF 5		DRAWING NUMBER
cont'd	Drive Assy, Gun and components				See Below
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	2-Pin (Dwg. B135B6695)				
	Examination for Defects	5		3.2	4.4.2.14
	Hardness	(a)		3.2	4.5.2
	Salt Spray Test	(a)		3.2	4.5.1
	Housing Drive (Dwg. F189F727)				
	Examination for Defects	5		3.2	4.4.2.15
	Salt Spray Test	(a)		3.2	4.5.1
	Bearing, Needle (Dwg. B135B5970-4)				
	Examination for Defects	5		3.2	4.4.2.16
	Hardness	(a)		3.2	4.5.2
	Motor, Elec Drive (Dwg. D905D490)				
	Examination for Defects	5		3.2	4.4.2.17
	Salt Spray Test	(a)		3.2	4.5.6
	Electrical Characteristic (Sht 1, Note 1 on dwg.)	(a)		MIL-M-8609	Certification
	Mechanical Characteristic (Sht 1, Note 9 on dwg.)	(a)		3.2	Certification
	Motor performance (Sht 1, notes 11 & 12)	(a)		3.2	Certification

NOTES: (a) This test shall be performed on the 5 samples subjected to the examination for defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 5 OF 5		DRAWING NUMBER
cont'd	Drive Assy, Gun and components				See Below
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Drive Assy, Gun (Dwg. F189F784)				
	Examination for Defects	5		3.2	4.4.2.19
	Performance characteristics	(a)		3.3.1	4.5.6
	Interchangeability	3		3.5	4.6

NOTES: (a) This test shall be performed on the 5 samples subjected to the examination for defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with the requirements.

4.4.2 Examination. The examinations listed in this specification for Classification of Defects shall be performed on inspection lots as defined in 4.4.1 of this specification.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Tests Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER D101D4498
4.4.2.1	Gearshaft				NEXT HIGHER ASSEMBLY F189F734
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Magnetic Particle		100%	3.2	4.5.3
Major					
101	Salt Spray Test	1		3.2	4.5.1
102	Test load on welded joint	(a)		3.2	4.5.4
103	Weld incomplete or inadequate		100%	3.2	Visual/Manual
104	Hardness		0.65%	3.2	4.5.2
105	Gearshaft diameter		0.65%	3.2	Gage
106	Run out of pitch diameter (surfaces A & B)		0.65%	3.2	Gage
107	Gear, outside diameter (2.613)		0.65%	3.2	Gage
108	Pitch diameter of spline		0.65%	3.2	Gage
109	Small bearing diameter		0.65%	3.2	Gage
110	Bearing diameter (.6252)		0.65%	3.2	Gage
111	Magnetic particle inspection mark missing		0.65%	3.2	Visual
Minor					
201	Height of view A		1.50%	3.2	Gage
202	Height of view C		1.50%	3.2	Gage
203	Marking misleading, illegible or missing		1.50%	3.2	Visual
204	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES: (a) This test load shall be performed on the same sample subjected to the Salt Spray Test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Gear, Spur		G116C3126
			NEXT HIGHER ASSEMBLY F189F734

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Magnetic Particle		100%	3.2	4.5.3
Major					
101	Salt Spray Test	1		3.2	4.5.1
102	Hardness		0.65%	3.2	4.5.2
103	Inside diameter (1.2450)		0.65%	3.2	Gage
104	Magnetic Particle Inspection mark missing		0.65%	3.2	Visual
Minor					
201	Width (.754)		1.50%	3.2	Gage
202	Marking misleading, illegible or missing		1.50%	3.2	Visual
203	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2020-10-26T04:27Z
Please see DLA Privacy and Security Notice and Terms of Use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Cover, End		D117D2954
			NEXT HIGHER ASSEMBLY F189F734

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Salt Spray Test	1		3.2	4.5.1
Minor					
201	Diameter of hole (E surface) (2 places) (0.252)		1.50%	3.2	Gage
202	Diameter of hole (2 places) (.103)		1.50%	3.2	Gage
203	Diameter (section A-A) (2.7525)		1.50%	3.2	Gage
204	Surface roughness (63)		1.50%	3.2	Gage
205	Marking misleading, illegible or missing		1.50%	3.2	Visual
206	Casting workmanship unsatisfactory (See MIL-C-6021)		1.50%	3.2	Visual
207	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER C11830451-1
4.4.2.4	Bearing, Ball Annular				NEXT HIGHER ASSEMBLY F189F734
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardness		0.65%	3.2	4.5.2
102	Grease missing		0.65%	3.2	Visual
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Reproduction may not modify this image, is the current version, and is for use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Bearing, Ball Annular	1 OF 1	C11830451-3
			NEXT HIGHER ASSEMBLY: F189F734

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardness		0.65%	3.2	4.5.2
102	Grease missing		0.65%	3.2	Visual
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Gear, Spur				C125C2691-1
					NEXT HIGHER ASSEMBLY
					C125C2694
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Magnetic Particle		100%	3.2	4.5.3
Major					
101	Salt Spray Test	1		3.2	4.5.1
102	Hardness		0.65%	3.2	4.5.2
103	Major Diameter on gear		0.65%	3.2	Gage
104	Minor diameter on spline		0.65%	3.2	Gage
105	Lubrication improper		0.65%	3.2	Visual
106	Magnetic particle inspection mark missing		0.65%	3.2	Visual
Minor					
201	Outside diameter		1.50%	3.2	Gage
202	Marking misleading, illegible or missing		1.50%	3.2	Visual
203	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER C125C2692
4.4.2.7	Gear, Spur				NEXT HIGHER ASSEMBLY C125C2694
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Magnetic Particle		100%	3.2	4.5.3
Major					
101	Salt Spray Test	1		3.2	4.5.1
102	Hardness		0.65%	3.2	4.5.2
103	Largest diameter		0.65%	3.2	Gage
104	Second largest diameter		0.65%	3.2	Gage
105	Width		0.65%	3.2	Gage
106	Magnetic particle inspection mark missing		0.65%	3.2	Visual
Minor					
201	Surface roughness (63)		1.50%	3.2	Visual-Measure
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Gear, Pinion		C125C2729
			NEXT HIGHER ASSEMBLY
			F189F734

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Magnetic Particle		100%	3.2	4.5.3
Major					
101	Salt Spray Test	1		3.2	4.5.1
102	Hardness		0.65%	3.2	4.5.2
103	Largest diameter		0.65%	3.2	Gage
104	Hole diameter (.1285)		0.65%	3.2	Gage
105	Pitch diameter		0.65%	3.2	Gage
106	Minor diameter		0.65%	3.2	Gage
107	Magnetic Particle Inspection Mark missing		0.65%	3.2	Visual
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Please review the above to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER B135B5773-25
4.4.2.9	Spacer, Sleeve				NEXT HIGHER ASSEMBLY F189F734
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Salt Spray Test	1		3.2	4.5.1
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Spacer, Sleeve				B135B5773-26
					NEXT HIGHER ASSEMBLY F189F734
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Salt Spray Test	1		3.2	4.5.1
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

Retrieved from http://assist.dla.mil - Document created: 2026-01-20 20:47:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.11	Spacer, Sleeve		B135B5773-27
			NEXT HIGHER ASSEMBLY
			F189F734

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Salt Spray Test	1		3.2	4.5.1
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

MIL-D-63516(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Shaft				B135B5783
					NEXT HIGHER ASSEMBLY
					F189F734
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Salt Spray Test	1		3.2	4.5.1
102	Hardness		0.65%	3.2	4.5.2
103	Length		0.65%	3.2	Gage
104	Diameter		0.65%	3.2	Gage
105	Surface roughness (16)		0.65%	3.2	Gage
Minor					
201	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Retainer	1 OF 1	B135B5785
			NEXT HIGHER ASSEMBLY F189F734

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Salt Spray Test	1		3.2	4.5.1
Minor					
201	True position of holes		1.50%	3.2	Gage
202	Marking misleading, illegible or missing		1.50%	3.2	Visual
203	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Pin				B135B6695
					NEXT HIGHER ASSEMBLY C125C2694
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Salt Spray Test	1		3.2	4.5.1
102	Hardness		0.65%	3.2	4.5.2
103	Diameter		0.65%	3.2	Gage
Minor					
201	Surface roughness (32)		1.50%	3.2	Visual-Measure
202	End chamfer missing		1.50%	3.2	Visual
203	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Housing, Drive	1 OF 1	F189F727
			NEXT HIGHER ASSEMBLY F189F734

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Salt Spray Test	1		3.2	4.5.1
102	Bottom outside diameter (1.0000)		0.65%	3.2	Gage
103	True position of holes		0.65%	3.2	Gage
104	Orientation of surfaces E and F (See Note 27 on dwg.)		0.65%	3.2	Gage
Minor					
201	Width of cavity (1.940)		1.50%	3.2	Gage
202	Diameter of hole (2 places) (.252)		1.50%	3.2	Gage
203	Surface roughness (63)		1.50%	3.2	Visual-Measure
204	Marking misleading, illegible or missing		1.50%	3.2	Visual
205	Poor casting workmanship (See MIL-C-6021)		1.50%	3.2	Visual
206	Notched tangs not removed (Note 24 on Dwg.)		1.50%	3.2	Visual
207	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER B135B5970-4
4.4.2.16	Bearing, Needle				NEXT HIGHER ASSEMBLY C125C2964
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardness		0.65%	3.2	4.5.2
102	Lubricant missing		0.65%	3.2	Visual
Minor					
201	Part marking illegible, improper or missing		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.17	Motor, Elec, Drive				D905D490 (2 shts)
					NEXT HIGHER ASSEMBLY F189F734
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Special sampling Connector bracket assy force test (See Note 8 on Dwg) (Sht 1)	Note 1			SMTE
Major					
101	Salt Spray			3.2	Certification
102	Motor requirements (See Note 1 on Dwg., excluding electrical characteristics) (Sht 1)			3.2	Certification
103	Electrical Characteristics (See Note 1.2 on Dwg.) (Sht 1)			3.2	SMTE
104	Mechanical Characteristics (Sht 1) (Note 9 on Dwg.)			3.2	Certification
105	Motor performance (see notes 11 & 12) (Sht 1)		100%	3.2	SMTE
106	Motor not totally enclosed		0.65%	3.2	Visual
107	Direction of rotation improper		0.65%	3.2	Visual
108	End plates (2) not pinned to field housing (Note 13 on Dwg.)		0.65%	3.2	Visual

NOTES:
1. Ten (10) motors shall be selected from each lot for test. If any motor fails to comply to the specified requirement the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
cont'd 4.4.2.17	Motor, Elec, Drive	2 OF 2	D905D490 (2 shts)
			NEXT HIGHER ASSEMBLY F189F734

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Weight, max		1.50%	3.2	Balance
202	Width of connector bracket, max (1.80) (Sht 2)		1.50%	3.2	Gage
203	Spline runout, max (.25) (Sht 2)		1.50%	3.2	Gage
204	Length from face of connector bracket to end of motor (4.08) (Sht 2)		1.50%	3.2	Gage
205	Diameter of surface B (2.752) (Sht 2)		1.50%	3.2	Gage
206	Length, max (6.45) (Sht 2)		1.50%	3.2	Gage
207	Height to center of shaft, max (3.62) (Sht 2)		1.50%	3.2	Gage
208	Notched tangs not removed (after installing inserts) (Note 7 on Dwg.) (Sht 1)		1.50%	3.2	Visual
209	Screw head not staked or safety wired, missing or improper (as applicable)		1.50%	3.2	Visual
210	Part missing, damaged or improperly assembled		1.50%	3.2	Visual
211	Marking misleading, illegible or missing, including damage to name plate		1.50%	3.2	Visual
212	Solder workmanship unsatisfactory (See MIL-W-45743)		1.50%	3.2	Visual
213	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Gear, Cluster, Matched	1 OF 1	C125C2694
			NEXT HIGHER ASSEMBLY: F189F734

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of holes (2)		0.65%	3.2	Gage
102	Any part missing, damaged or improperly assembled		0.65%	3.2	Visual
103	Grease missing		0.65%	3.2	Visual
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER F189F734
4.4.2.19	Drive Assy, Gun				NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Performance characteristics	2	0-1	3.3.1	4.5.6
102	Interchangeability	3		3.5	4.6
103	Any part missing, damaged or improperly assembled		0.65%	3.2	Visual
104	Grease missing		0.65%	3.2	Visual
Minor					
201	Axial plate of gearshaft (see Note 5 on dwg.)		1.50%	3.2	Gage
202	Marking misleading, illegible or missing		1.50%	3.2	Visual
203	Identification plate missing		1.50%	3.2	Visual
204	Safety wire improperly twisted or missing		1.50%	3.2	Visual
205	Touch-up paint missing (where required)		1.50%	3.2	Visual
206	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63516(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.20	Drive Assy Gun (Packaging Data Sheet)	1 OF 1	PSD P189F734
			NEXT HIGHER ASSEMBLY ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Waterproofness of sealed barrier bag destroyed by separation of heat seal, tear or puncture		0.65%	3.2	Visual
102	Box damaged to the extent that contents are exposed		0.65%	3.2	Visual
Minor					
201	Marking missing, misleading or unidentifiable		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (See 4.5). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.3 herein.

4.5 Test method and procedures.

4.5.1 Salt spray test. Samples shall be selected from each protective finished lot and subjected to the salt spray test in accordance with the procedures specified in ASTM B 117. Unless otherwise specified on the applicable drawing the minimum exposure time shall be one and one half (1 1/2) hours using a 5% sodium chloride solution. Rejection shall be in accordance with the applicable protective finish specification.

4.5.2 Hardness. Samples shall be selected from each heat treatment batch (see NOTE) and subjected to this test. If any part fails to comply with the requirement specified on the applicable drawing, the part shall be classed defective. The hardness test shall be performed in accordance with the procedure specified in ASTM E 18. Each heat treat batch shall remain segregated until completion of all hardness tests.

NOTE: A heat-treatment batch shall be defined as parts that have been heat-treated at the same time in the same furnace and quench batch for each of all phases of the entire heat-treatment process.

4.5.3 Magnetic particle test. The procedures and equipment for performing this inspection shall be in accordance with MIL-I-6868 and MIL-W-63150 and shall be satisfactory to the Contracting Officer. Personnel performing this inspection shall be qualified in accordance with MIL-STD-410.

4.5.4 Test load on welded joint. The welded joint shall be subjected to an axial test load as specified on Dwg. D101D4498. Any gearshaft which fails to comply with the specified compression or extension requirements shall be classed defective. Test equipment and procedures shall be satisfactory to the contracting officer. (See 4.4.4)

4.5.5 Bearing stability (First Article only). The annular ball bearing (Dwg. C11830451) shall be subject to the stability test for the temperature range specifed. If any bearing fails to comply with the specified requirement it shall be classed defective.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.5.6 Performance characteristics. The gun drive assembly shall be tested for compliance with the specification requirements of acceleration, steady-state operation, and maximum torque. Any assembly which fails to comply with the specified performance characteristic requirements or evidences degradation of its operational performance shall be classed defective. Test equipment and procedures shall be satisfactory to the contracting officer. (See 4.4.4)

4.5.6.1 Environmental characteristics (first article inspection only).

4.5.6.1.1 Temperature-altitude. Drive assemblies shall be subjected to the temperature-altitude test specified in MIL-STD-810 to determine compliance with the applicable specification requirements. The drive assemblies shall be function tested for compliance with the applicable specification requirements of acceleration, steady-state operation and maximum torque when the requirements specify that the system shall be operated.

4.5.6.1.2 Vibration test. Drive assemblies shall be vibration tested for compliance with the applicable specification requirement for 400 duty cycles without load. While operating as specified, the drive motor shall be vibrated at 80 Hz with a resultant magnitude and direction derived from a .15 inch total displacement parallel to the drive rotor axis and .40 inch total displacement perpendicular to the drive rotor axis. After completion of the 400 duty cycles, the drive assembly (motor stopped) shall be vibrated at 80 Hz and .40 inch total displacement parallel to the drive rotor axis for one hour. Drive assemblies shall be mounted to a vibration table using the mounting techniques specified in MIL-STD-810, Method 514.1. The general requirements of MIL-STD-810 shall apply as applicable. Pre-test and after-test performance data shall be determined by function testing as specified in the applicable specification requirements of acceleration, steady-state operation, and maximum torque. Any drive assembly which fails to comply with the specified functioning test shall be classed defective.

4.5.6.2 Proof test (first article inspection only). Five (5) gun drive assemblies shall be forwarded to the cite specified by the contracting agency and assembled to an accepted Universal Turret Subsystem (UTS). The gun drive assembly shall function in accordance to the intent of the design and with no evidence of malfunction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.6 Interchangeability testing.

4.6.1 Gun drive assembly. A sample of 3 drive assemblies, randomly selected from each inspection lot, shall be tested for interchangeaiblity using the test method specified in 4.6.3. Drive assemblies taken for interchangeability testing shall have been found satisfactory in all other examinations and tests. Test frequency may be reduced to not less than one test of 3 drive assemblies each lot when a record of consistently satisfactory results has been established. The 3 drive assemblies shall comply with the applicable specification functioning requirements using the procedures specified in the applicable specification requirements of acceleration, steady-state operation and maximum torque, before and after the interchange of parts. It shall be the responsibility of the agency responsible for the test to inform the contracting officer by letter or other methods satisfactory to the contracting officer, when selective assembly has become necessary. Statement shall contain as a minimum, the contractor's name and FSCM, contract number, parts involved in the selective fit, conditions of parts and rationale for action taken. No hand refinement of parts shall be permissible without approval by the contracting officer. No malfunctions shall be allowed in the functioning tests. Failure of drive assembly in the interchangeability test shall cause retest or rejection of the represented lot. At the discretion of the Government representative, an interchangeability retest may be allowed without reconditioning the lot. Failure during the retest shall cause rejection of the represented lot subject to reconditioning and further test as a reconditioned lot. A sample of 6 drive assemblies from each retest or reconditioned lot shall be tested using the same procedure described above.

4.6.2 Repair parts. At least 2 parts from each inspection lot of concurrent repair parts shall be subjected to the interchangeability test specified in 4.6.3. Failure of any part to meet the interchangeability requirements shall be cause for rejection of the represented lot, subject to reconditioning and further test as a reconditioned lot. A sample consisting of double the number of parts used in the original test shall be tested from each reconditioned lot using the test methods specified above.

4.6.3 Test method for interchangeability. Drive assemblies shall be tested by disassembling and then reassembling the parts specified below. Parts shall be disassembled from the drives and identical parts placed together and mixed. Any commercial part (screws, pins, etc) rendered unserviceable by disassembly shall be replaced without penalty to the interchangeability test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

LIST OF INTERCHANGEABLE PARTS

Gear, Cluster, Matched	(C125C2694)
Cover, End	(D117D2954)
Gearshaft	(D101D4498)
Gear, Pinion	(C125C2729)
Shaft	(B135B5783)
Retainer	(B135B5785)
Housing, Drive	(F189F727)

4.6.4 Repair parts. Concurrent repair parts shall be tested by disassembling 2 drive assemblies previously tested in 4.6.3 and then reassembling them using the concurrent repair parts. No hand refinement of parts will be allowed and the assembled drive assemblies shall operate and function as intended. This test may be performed independently of the drive assembly interchangeability testing specified in 4.6.1 and at more frequent intervals using accepted drive assemblies taken from current production.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. The Gun Drive Assemblies shall be prepared, packaged, packed and marked in accordance with Packaging Data Sheet (PDS) 189F734 for the level of protection specified in the contract. (See 6.2)

6. NOTES

6.1 Intended use. The Gun Drive Assembly covered by this specification is intended for use with Gun, automatic, 20MM: M197 on Armament Subsystem, Helicopter, 20MM: M97E1 and M97E2.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Serialization requirements, if applicable.

d. Quality Conformance Inspection, if other than specified in Section 4 of this specification.

e. First Article sample requirements, if other than specified in Section 4 of this specification.

f. Packaging requirements, if other than specified in Section 5 of this specification.

g. Certificates of conformance for each lot or shipment of product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

6.2.2 Contract data requirements. Contract data requirements for inspection equipment design shall conform to Data Item Description DI-R-1714, Tailored.

6.3 Submission of inspection equipment design for approval. See MIL-W-63150. Equipment designs shall be submitted as required to: Commander, ARRADCOM, ATTN: DRDAR-QAF-I, Dover, NJ 07801. Request letter of submittal should state contractor, contract number, specification number, item nomenclature, and classification of defects and test paragraphs.

6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to: Commander, ARRADCOM, ATTN: DRDAR-QAF-S, Dover, NJ 07801.

Custodian:
Army-AR

Preparing activity:
Army-AR

Project 1005-A610

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.